

Rebecca Kyer

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Education

MICHIGAN STATE UNIVERSITY

Ph.D. Candidate; thesis advisor: Jay Strader

M.S., Astrophysics and Astronomy

East Lansing, MI

Nov 2024 – present

Aug 2022 – Nov 2024

UNIVERSITY OF WASHINGTON

B.S., Physics and Astronomy

Astronomy Departmental Honors

Seattle, WA

Jun 2021

Past Research Appointments

GREEN BANK OBSERVATORY

Graduate Student in Residence

Green Bank, WV

May 2026

Publications

Refereed.....

6. “Three New Likely Spider Millisecond Pulsar Binaries and Optical Kinematic Tracers of Intrabinary Shocks” Strader, J., **Kyer, R.**, Chomiuk, L., et al. submitted to MNRAS
5. “Investigating Four New Candidate Redback Pulsars Discovered in the Image Plane.” Petrou, F., Hurley-Walker, N., McSweeney, S. et al. (incl. **Kyer, R.**) 2025, PASA, 42, 139
4. “Multiwavelength Evidence for Two New Candidate Transitional Millisecond Pulsars in the Subluminous Disk State: 4FGL J0639.1–8009 and 4FGL J1824.2+1231.” **Kyer, R.**, Roy, S., Strader, J., et al. 2025, ApJ, 983, 112
3. “A Survey for Radio Emission from White Dwarfs in the VLA Sky Survey.” Pelisoli, I., Chomiuk, L., Strader, J., et al. (incl. **Kyer, R.**) 2024, MNRAS, 531, 1805
2. “Monitoring the X-ray Variability of Bright X-ray Sources in M33.” **Kyer, R.**, Albrecht, S., Williams, B. F., et al. 2024, ApJ, 961, 168
1. “Revisiting the Classics: On the Evolutionary Origin of the ‘Fe II’ and ‘He/N’ Spectral Classes of Novae.” Aydi, E., Chomiuk, L., Strader, J. et al. (incl. **Kyer, R.**) 2024, MNRAS, 527, 9303

Non-refereed.....

“The Advanced X-ray Imaging Satellite Community Science Book.” Koss, M., Aftab, N., Allen, S. et al (incl. **Kyer, R.**) 2025, arXiv:2511.00253

“Unlocking the Rapid X-ray Variability of Transitional Millisecond Pulsars.” **Kyer, R.**, Ng, M., Coti Zelati, F. et al.

“Catching Spiders with AXIS.” Coti Zelati, F., An, H., Baglio, M. C. et al. (incl. **Kyer, R.**)

Transient Name Server Classification Report No. 2024-2571, classification of SN 2024pxg as a SN II. Bostroem, K. A., **Kyer, R.**, Strader, J. et al. 2024

Classification reports in The Astronomer’s Telegram (No. 16583, 16440, 16294, 17878)

Telescope Time

XMM-Newton AO-25 ID 098076, 35 ksec awarded, Scientific PI	2025
Green Bank Telescope 2026A GBT26A-226, 15 hr awarded/4.5 hr observed, PI/Observer	2025
Gemini-North Fast Turnaround ID GN-2025B-FT-202, 0.38 hr, PI	Aug 2025
Green Bank Telescope 2025B Legacy ID QS402, 15 hr, Co-I/Observer	2025
XMM-Newton AO-24 ID 096163, 57 ksec awarded, Scientific PI	2025
Swift Target of Opportunity ID 21032, 8.5 ksec, PI	Aug 2024
Swift Target of Opportunity ID 20572, 3.8 ksec, PI	June 2024
200+ hours remote observing with the 4.1-m SOAR telescope	2023–present

Research Advising Experience for MSU Undergraduate Students

Jadyn Elster *May 2026–present*

Learning to analyze optical spectra from long-term monitoring of an evolved massive star with a hidden compact object companion. Co-author of a manuscript in preparation on the results of this monitoring data.

Andrew Harms, co-advised with Ryan Urquhart *May 2025–present*

Learned to combine archival X-ray and radio data to constrain the prevalence of hundreds-of-pc-scale radio/optical nebulae associated with ultraluminous X-ray sources in the Southern Hemisphere sky to predict yield from future deep radio surveys. Made an independent discovery of a new nebula and learned to analyze a follow-up optical spectrum I obtained with the Gemini Fast-Turnaround program. Preparing manuscript for first-author publication of both results.

Subhroja Roy (now U. Notre Dame Nuclear Astrophysics Ph.D. student) *May 2024–May 2025*

Learned to combine archival γ -ray, X-ray and optical data sets to identify several potential spider MSPs and analyze follow-up optical spectroscopy. Completed astronomy senior thesis presenting two new strong candidate binary MSPs, one of which has been published with Subhroja as second author and the other is to be included in an upcoming publication.

Research Presentations

University of Tasmania Physics & Astronomy Journal Club talk (Hobart, Australia)	Aug 2026
HEAD HMXBs and Pulsar Phenomena talk (Pasadena, CA)	June 2026
AAS 248 Neutron Stars and Pulsar Physics dissertation talk (Pasadena, CA)	June 2026
AAS 248 Science of Stellar Remnants press conference (Pasadena, CA)	June 2026
Green Bank Observatory (GBO) Science Lunch talk (Green Bank, WV)	May 2026
GBO/NRAO Radio Astronomy Bootcamp science talk (Green Bank, WV)	May 2026
MSU Astronomy Seminar talk (East Lansing, MI)	Jan 2026
AAS 247 AXIS Special Session invited talk (Phoenix, AZ)	Jan 2026
AAS 247 Accretion and Evolution in X-ray Binary Systems talk (Phoenix, AZ)	Jan 2026
Compact Objects in Michigan and Ontario (COMO) talk (Flint, MI)	Sept 2025

Vasto Accretion Meeting talk (Vasto, Italy)	<i>June 2025</i>
AXIS Community Science Meeting talk (Annapolis, MD)	<i>May 2025</i>
Int'l Symposium of Science with the SOAR Telescope talk (East Lansing, MI)	<i>May 2025</i>
ESAC XMM-Newton Science Workshop poster (Madrid, Spain)	<i>June 2024</i>
COMO talk (Dearborn, MI)	<i>May 2024</i>

Outreach Activities

Letters to a Pre-Scientist Penpal	<i>2 penpals</i>
Astronomy on Tap Speaker (Lansing, MI)	<i>Nov 2023, Apr 2025, July 2026</i>
Campus Observatory Public Night Volunteer (East Lansing, MI)	<i>Aug 2022–present</i>

Professional Service

COMO Conference Local Organizing Committee Chair	<i>2026</i>
ESA NewAthena Science Community Working Group Member	<i>Feb 2026–present</i>
AAS 247 Chambliss Poster Competition Judge	<i>Jan 2026</i>
NASA AXIS Probe CO/SNR Working Group Member	<i>Dec 2024–Mar 2026</i>
P-A Research Experiences for Drew Scholars Junior Director	<i>Apr 2024–present</i>
Faculty Search Committee Graduate Student Representative	<i>Feb 2024</i>
Stellar Mentorship Program Coordinator and Peer Mentor	<i>Aug 2023–present</i>
Astro Coffee Journal Club Organizer	<i>Jun 2023–present</i>

Teaching Experience

Teaching Assistant AST 208: Planets and Telescopes (30 students; MSU)	<i>Jan–May 2024</i>
Teaching Assistant ISP 205L: Visions of the Universe (400 students; MSU)	<i>Aug 2022–Dec 2023</i>
Instructor GEN ST 199: Astronomy First-year Interest Group (30 students; UW)	<i>Sep–Dec 2020</i>

Awards and Honors

Honorable Mention, NSF Graduate Research Fellowship	<i>Apr 2024</i>
Graduate Teaching Assistant Award, MSU Physics and Astronomy	<i>Apr 2024, Apr 2023</i>
Mervyn Mork Excellence in Teaching Award, MSU College of Natural Science	<i>Nov 2023</i>
Baer Prize for Excellence in Research, Academics and Outreach, UW Astronomy	<i>May 2021</i>